Scalability & Future Plan

Date	6 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Predictions depend on the quality of historical weather data.	May reduce prediction accuracy.	High
2	Supports limited concurrent users.	Performance may decrease with high traffic.	Medium
3	No real-time weather API integration.	Predictions require manual data entry.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users.	Deploy on cloud with load balancing to support thousands of users.
2	Data Storage	Local dataset storage.	Use cloud databases for scalable and secure storage.
3	Performance	Basic Random Forest prediction.	Optimise model and implement caching for faster responses.
4	Security	Basic Flask application.	Add user authentication, HTTPS and secure API access.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time weather API integration	2 Months	More accurate predictions
Phase 2	SMS & Email flood alerts	4 Months	Faster emergency notifications
Phase 3	Mobile application (Android/iOS)	6 Months	Improved accessibility
Phase 4	AI-based flood risk visualisation using GIS maps	8 Months	Better decision-making and disaster management